

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T49 · Sine Rule, Cosine Rule and Area of a Triangle

With Answers

Questions with full worked solutions

■ 25 pages

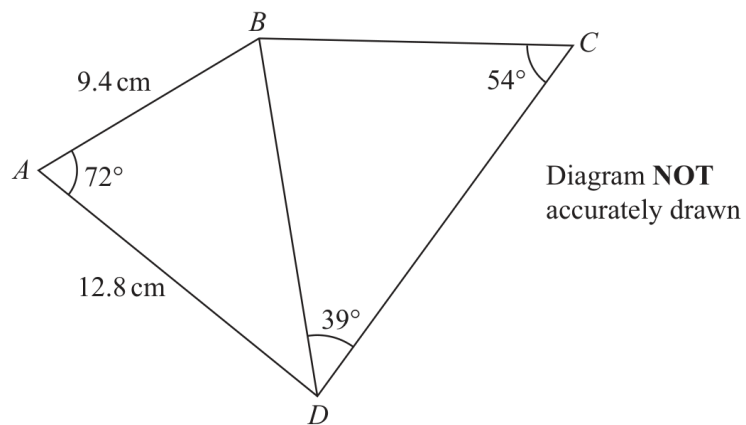
FAMILY MAP

Sine Cosine Rule Area of Triangles

F01	Resolve mixed non-right-triangle chains	6 Q 31 marks
F02	Use the sine rule	3 Q 16 marks
F03	Use the cosine rule	1 Q 3 marks
F04	Use area equals one-half $ab \sin C$	1 Q 3 marks

Working Unit 13

4MA1-1H Q20

ORIGINAL QUESTION**5 marks****20**

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Worked Solution 13

Cosine rule followed by sine rule

UNIT 1
5 marks**Find the shared side**

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ, \quad BD = 13.335597\dots$$

Use the sine rule

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ} \Rightarrow BC = \frac{13.335597\dots \sin 39^\circ}{\sin 54^\circ} = 10.3735\dots$$

Final answer

$$BC = 10.4 \text{ cm (3 s.f.)}$$

Dependency check Retain the unrounded value of BD for the second triangle.

Question 13

4MA1-1H Q19

ORIGINAL QUESTION

5 marks

19 Here is quadrilateral $ABCD$

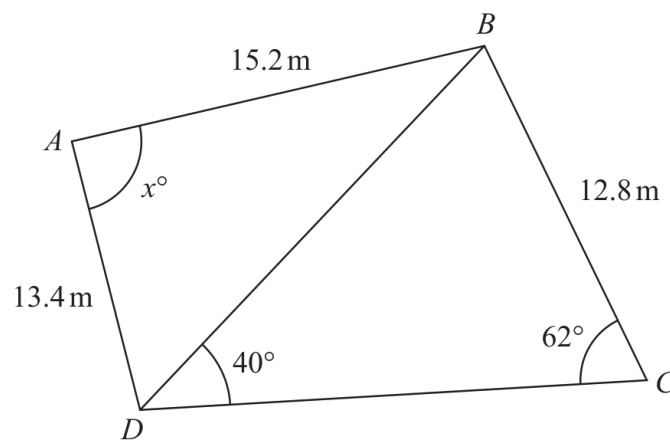


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

Worked Solution 13

Sine rule and cosine rule

MODULAR UNIT 1

5 marks

Diagonal

$$BD = 12.8 \sin 62^\circ / \sin 40^\circ = 17.582 \dots$$

Required angle

$$\cos x = \frac{13.4^2 + 15.2^2 - BD^2}{2(13.4)(15.2)}.$$

Final answer

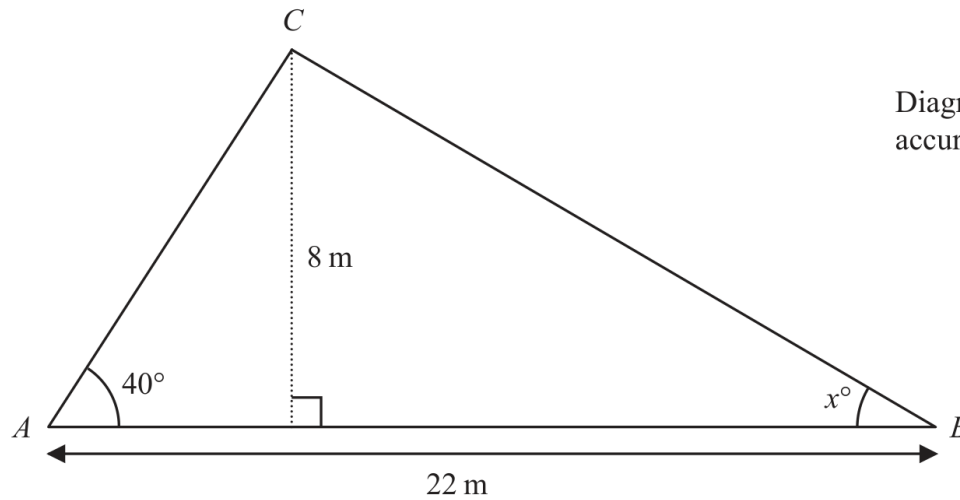
$$75.6^\circ$$

Question 09

4MA1-1H Q10 demand

ORIGINAL QUESTION UNIT 1

5 marks

10 Here is triangle ABC Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

 $x = \dots\dots\dots$

Worked solution 09

Angle in a triangle from a perpendicular height

MODULAR UNIT 1

5 marks

Demand – Angle in a triangle from a perpendicular height

Find the left base segment

$$AD = \frac{8}{\tan 40^\circ} = 9.534\dots$$

Find the right base segment

$$DB = 22 - 9.534\dots = 12.466\dots$$

Use the right triangle at B

$$\tan x = \frac{8}{12.466\dots}$$

Evaluate

$$x = 32.690\dots^\circ \approx 32.7^\circ$$

Final answer 32.7°

Question 26

4MA1-2H Q22

ORIGINAL QUESTION

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

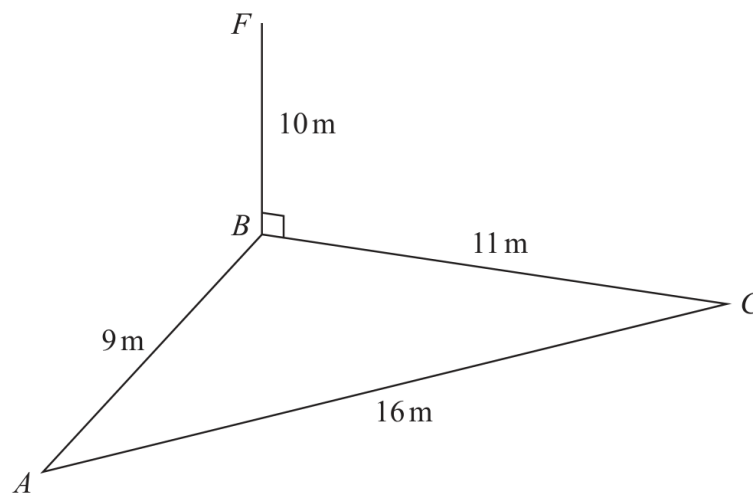


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

Worked Solution 26

Angle of elevation from a perpendicular foot

MODULAR UNIT 1

5 marks

Find the ground angle

$$\cos \angle BCA = \frac{11^2 + 16^2 - 9^2}{2(11)(16)} = \frac{37}{44} \Rightarrow \angle BCA \approx 32.763^\circ.$$

Find BD

$$BD = 11 \sin 32.763^\circ \approx 5.95 \text{ m.}$$

Use the vertical flagpole

$$\tan \angle FDB = \frac{BF}{BD} = \frac{10}{5.95}.$$

Evaluate

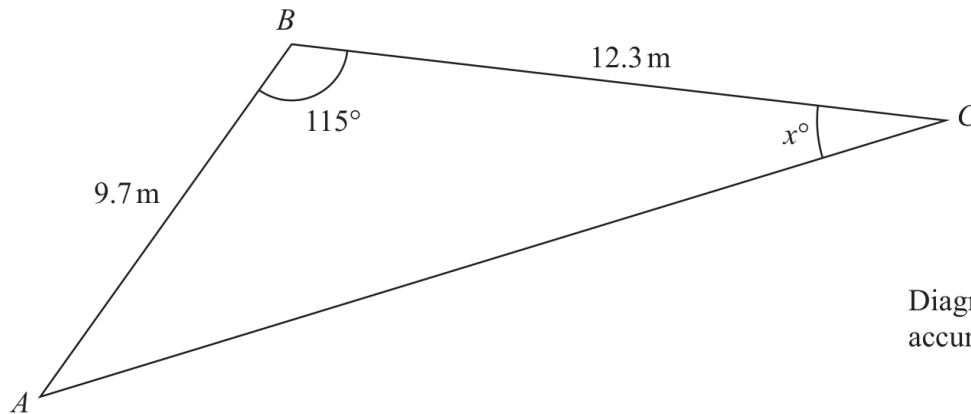
$$\angle FDB \approx 59.2^\circ.$$

Final answer

$$59.2^\circ$$

Original question 12

4MA1-1H Q18 Q18

ORIGINAL QUESTION**5 marks****18** Here is triangle ABC Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

 $x =$

Worked solution

Family: Use the sine rule • Variant: Find an acute angle using the sine rule

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 12 MODULAR UNIT 1
Chapter: Sine, Cosine Rule & Area of Triangles 5 marks
Q18 – Chapter: Sine, Cosine Rule & Area of Triangles
Family: Use the sine rule • Variant: Find an acute angle using the sine rule
Method
<i>By the cosine rule, $AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos(115\text{degrees})$.</i>
Why: The two known sides enclose the known 115-degree angle.
Method
$AC^2 = 346.2 \dots$, so $AC = 18.606 \dots m$.
Why: Evaluate the cosine-rule expression without premature rounding.
Method
Using the sine rule, $\sin(x)/9.7 = \sin(115 \text{ degrees})/18.606 \dots$
Why: Side AB = 9.7 is opposite angle x, and AC is opposite 115 degrees.
Method
Therefore $\sin(x) = 9.7 \sin(115 \text{ degrees})/18.606 \dots = 0.472 \dots$
Why: Rearrange the sine-rule equation for $\sin(x)$.
Method
$x = \sin^{-1}(0.472 \dots) = 28.2 \text{ degrees to 3 significant figures}$.
Why: Take the inverse sine in degree mode and round once at the end.
Final answer
angle: $x = 28.2 \text{ degrees}$

FINAL ANSWER

Original question 27

4MA1-2H Q23

ORIGINAL QUESTION**6 marks**

23 The diagram shows triangle PQR

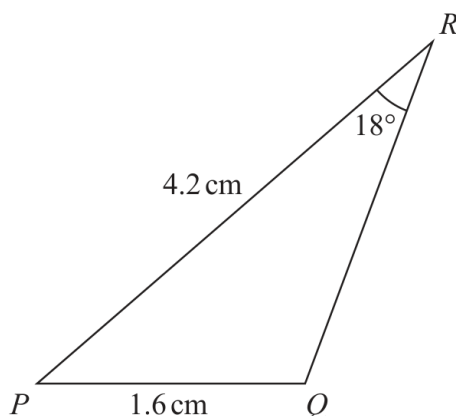


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Worked solution

Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 27 MODULAR UNIT 1
Chapter: Sine, Cosine Rule & Area of Triangles 6 marks
Q23 – Chapter: Sine, Cosine Rule & Area of Triangles
Family: Use the sine rule • Variant: Resolve the ambiguous sine-rule angle
Method
Using the sine rule, $\sin Q/4.2 = \sin 18^\circ/1.6$, giving $Q = 54.2^\circ$ or 125.8° .
Why: The sine equation has an acute and an obtuse possibility.
Method
The question says Q is obtuse, so $Q = 125.8^\circ$.
Why: Select the permitted branch.
Method
$P = 180 - 125.8 - 18 = 36.2^\circ$.
Why: Angles in a triangle total 180° .
Method
The corresponding side RQ is 3.0585... cm.
Why: Use the cosine rule or the sine rule after finding P.
Method
$\text{Area} = 1/2 \times 4.2 \times 1.6 \times \sin 36.2^\circ$.
Why: Use two sides and the included angle.
Method
$\text{Area} \approx 1.98 \text{ cm}^2$ (3 s.f.).
Why: Round the calculator value.
Final answer
$\text{area} : 1.98 \text{ cm}^2$

FINAL ANSWER

14 The diagram shows parallelogram $EFGH$.

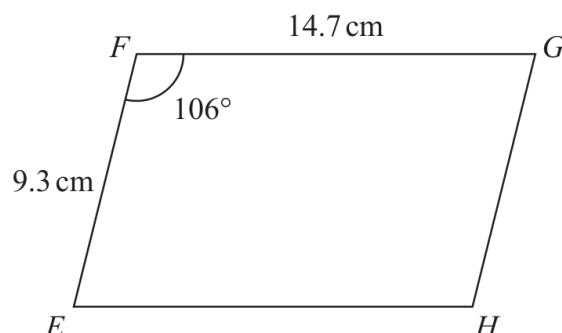


Diagram **NOT**
accurately drawn

$$EF = 9.3 \text{ cm}$$

$$FG = 14.7 \text{ cm}$$

$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

- (b) Work out the length of the diagonal EG of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

Worked solution

January 2020 | Question 14 | 3 marks

Family: Use the cosine rule | Variant: Use cosine rule inside a quadrilateral or composite shape

METHOD / WORKING

$$1. EG^2 = 9.3^2 + 14.7^2 - 2(9.3)(14.7)\cos 106^\circ.$$

$$2. EG = 19.440\dots \text{ cm.}$$

$$3. \text{ To 3 s.f., } EG = 19.4 \text{ cm.}$$

FINAL ANSWER

19.4 cm

18

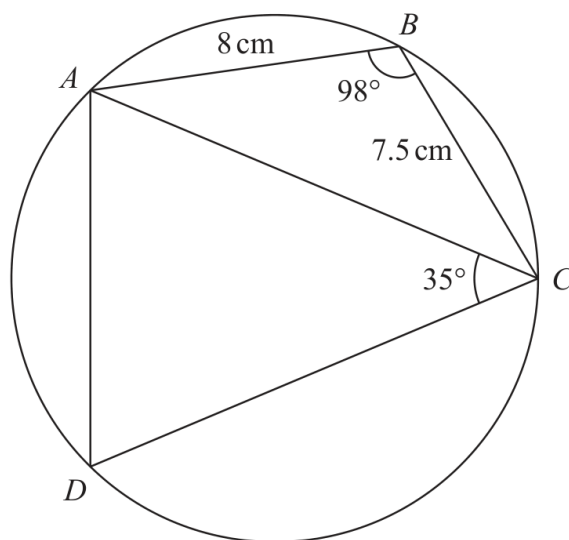


Diagram **NOT**
accurately drawn

$ABCD$ is a quadrilateral where A , B , C and D are points on a circle.

$$AB = 8 \text{ cm}$$

$$BC = 7.5 \text{ cm}$$

$$\text{Angle } ABC = 98^\circ$$

$$\text{Angle } ACD = 35^\circ$$

Work out the perimeter of quadrilateral $ABCD$.

Give your answer correct to one decimal place.

..... cm

Worked solution

June 2021 | Question 18 | 6 marks

Family: Resolve mixed non-right-triangle chains • Variant: Combine non-right trigonometry with circle or elevation geometry

Method / working

1. Cyclic quadrilateral: $\angle ADC = 180 - 98 = 82^\circ$.
2. In triangle ADC, $\angle DAC = 180 - 82 - 35 = 63^\circ$.
3. Use the cosine rule in triangle ABC to find $AC \sim 11.7$ cm.
4. Use the sine rule: $AD = AC \cdot \sin 35^\circ / \sin 82^\circ \sim 6.77$ cm.
5. Use the sine rule: $DC = AC \cdot \sin 63^\circ / \sin 82^\circ \sim 10.5$ cm, then add $8 + 7.5 + AD + DC$.
6. Round the perimeter to 1 decimal place: 32.8 cm.

Final answer: 32.8 cm

18 Here is a quadrilateral $ABCD$.

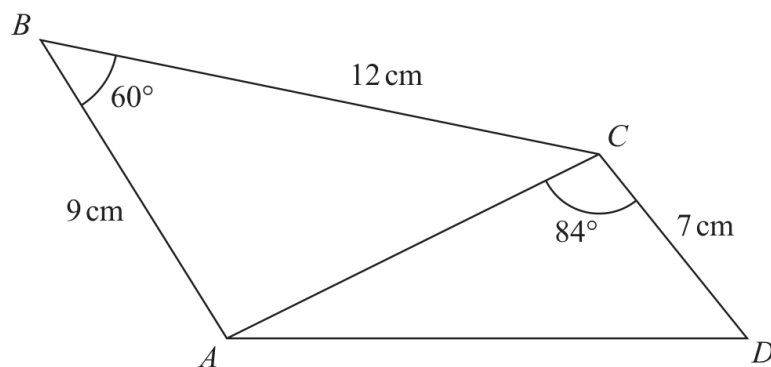


Diagram **NOT**
accurately drawn

Calculate the area of quadrilateral $ABCD$.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2

Worked solution

November 2021 | Question 18 | 5 marks

Family: Resolve mixed non-right-triangle chains | Variant: Use two non-right triangles sharing a diagonal

METHOD / WORKING

1. Area of triangle ABC :
2. $\frac{1}{2}(9)(12)\sin 60^\circ = 46.765\dots$
3. Find AC using the cosine rule:
4. $AC^2 = 9^2 + 12^2 - 2(9)(12)\cos 60^\circ$
5. $AC = 10.816\dots$
6. Area of triangle ACD :
7. $\frac{1}{2}(10.816\dots)(7)\sin 84^\circ = 37.642\dots$
8. Total area:
9. $46.765\dots + 37.642\dots = 84.407\dots$
10. Answer: 84.4cm^2 .

FINAL ANSWER

 84.4cm^2 .

- 16 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

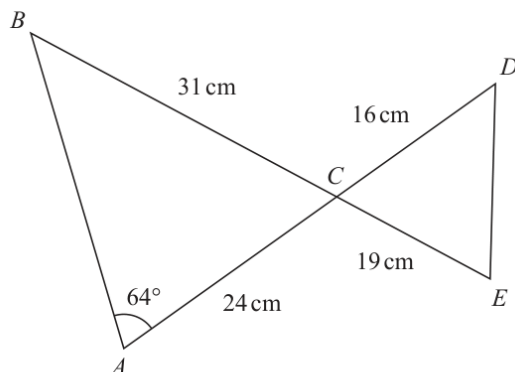


Diagram **NOT**
accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.



..... cm

(Total for Question 16 is 5 marks)

Answer reference | January 2023 | Question 16

Family: Use the sine rule • Variant: Find a side using the sine rule

16	$\frac{\sin ABC}{24} = \frac{\sin 64}{31}$ oe		5	M1
	$(ABC =) \sin^{-1}\left(\frac{24 \times \sin 64}{31}\right) (= 44. \dots)$			M1
	$180 - "44. \dots" - 64 (= 71.9 \dots)$			M1 accept 72
	$(DE^2 =) 16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"$ or $(DE =) \sqrt{16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"}$ or $(DE =) \sqrt{617 - 181.8 \dots}$ or $\sqrt{428.166 \dots}$			M1 for DE^2 or DE
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	20.7		A1 awrt 20.7
				Total 5 marks

Worked solution

January 2023 | Question 16 | 5 marks

Family: Use the sine rule • Variant: Find a side using the sine rule

Official final mark-scheme pages are embedded immediately after this question.

19 The diagram shows triangle ABC

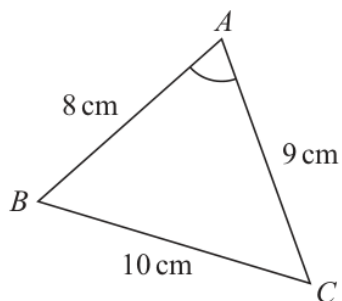


Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

(Total for Question 19 is 3 marks)

Worked solution

November 2025 | Question 19 | 3 marks

Family: Use area equals one-half ab sine C | Variant: Recover a side or angle from triangle area

METHOD / WORKING

Worked solution
November 2025 Question 19 3 marks
Family: Use area equals one-half ab sine C Variant: Recover a side or angle from triangle area
Solution
Use the cosine rule for angle BAC:
$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$
$\cos A = \frac{45}{144}$
$A = 71.790\dots^\circ$
Answer: 71.8° .
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER